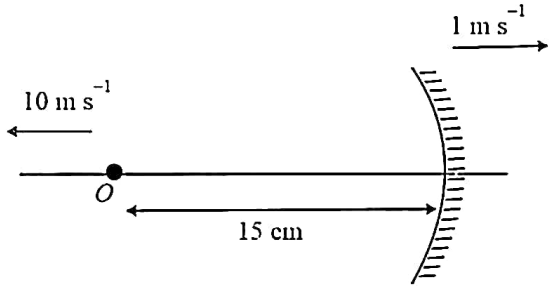
Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1 **(Only One Option Correct Type)**

A mirror of radius of curvature 20 cm and an object which is placed at a distance of 15 cm are both moving with velocities $1\ \mathrm{m\ s^{-1}}$ and $10\ \mathrm{m\ s^{-1}}$ as shown in diagram. The velocity of the image at this situation is $9v_P$. Find v_P.



- 4
- 5
- 8
- 10

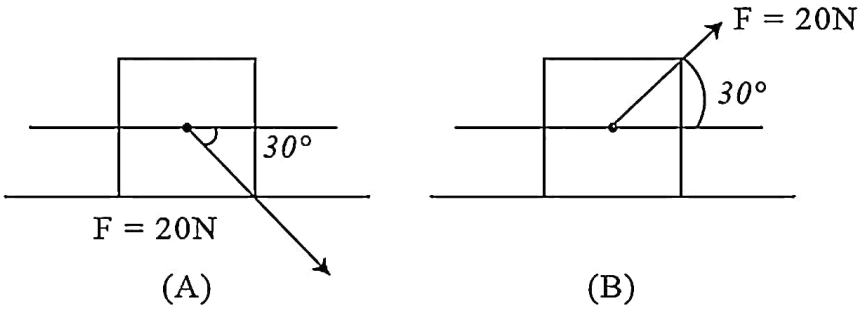
Question No. 2 **(Only One Option Correct Type)**

A parallel plate capacitor is charged and then disconnected from the source of potential difference. If the plates of the condenser are moved farther apart by the use of insulated handle, which of the following is true?

- The charge on the capacitor increases
- The charge on the capacitor decreases
- The capacitance of the capacitor increases
- The potential difference across the plates increases

Question No. 3 **(Only One Option Correct Type)**

Two cases shown in figure. A block of mass 5kg is (i) pushed in case (A) and (ii) pulled in case (B), by a force $\mathrm{F} = 20\mathrm{N}$, making an angle of 30° with the horizontal, as shown in the figures. The coefficient of friction between the block and floor is $\mu = 0.2$. what is the difference between the accelerations of the block, in case (B) and case (A) ? $\left(take\ g = 10ms^{-1}\right)$



- $0.8\mathrm{ms}^{-2}$
- $0\mathrm{ms}^{-2}$
- $3.2\mathrm{ms}^{-2}$

D) $0.4\ \mathrm{ms}^{-2}$

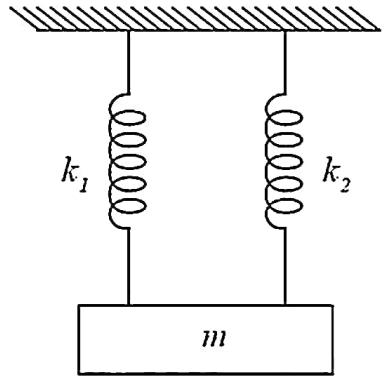
Question No. 4 **(Only One Option Correct Type)**

A long solenoid having 100 turns per cm carries a current of $\frac{4}{\pi}$ A. At the centre of it is placed a coil of 200 turns of cross-sectional area 25 cm^2 having its axis parallel to the field produced by the solenoid. When the direction of the current in the solenoid is reversed with in 0.04 s, the induced emf in the coil is....

- 0.2 V
- 0.4 V
- 0.002 V
- 0.016 V

Question No. 5 **(Only One Option Correct Type)**

A mass m is suspended separately by two different springs in successive order, then time periods is t_1 and t_2 respectively. If m is connected by both springs as shown in figure, then time period is t_0, the correct relation is



- $t_0^2 = t_1^2 + t_2^2$
- $t_0^{-2} = t_1^{-2} + t_2^{-2}$
- $t_0^{-1} = t_1^{-1} + t_2^{-1}$
- $t_0 = t_1 + t_2$

Question No. 6 **(Only One Option Correct Type)**

Gas escapes from the surface of a planet because it acquires an escape velocity. The escape velocity will depend on which of the following factors?
I. Mass of the planet.
II. Mass of the particle escaping.
III. Temperature of the planet.
IV. Radius of the planet.

Select the correct answer from the codes given below.

- I and II
- II and IV
- I and IV
- I, III and IV

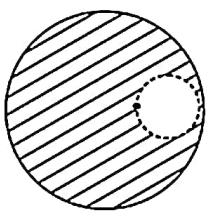
Question No. 7 **(Only One Option Correct Type)**

A light of wavelength 4000Ã . . . is incident on a metal having threshold frequency 5.5×10^{14} Hz. If light has intensity $\mathrm{I} = 100\ \mathrm{W/m^2}$ then the number of incident photons per $\mathrm{m^2}$ per second, is :-

- 1×10^{20}
- 2×10^{20}
- 3×10^{20}
- 4×10^{20}

Question No. 8 **(Only One Option Correct Type)**

From a solid sphere of mass M and radius R, a spherical portion of radius $\frac{\mathrm{R}}{2}$ is removed, as shown in the figure. Taking gravitational potential $\mathrm{V} = 0$ at $\mathrm{r} = \infty$ the potential at the centre of the cavity thus formed is
$(\mathrm{G} = \text{Universal gravitational constant})$.



- $\frac{-2GM}{3R}$
- $\frac{-2GM}{R}$
- $\frac{-GM}{2R}$
- $\frac{-GM}{R}$

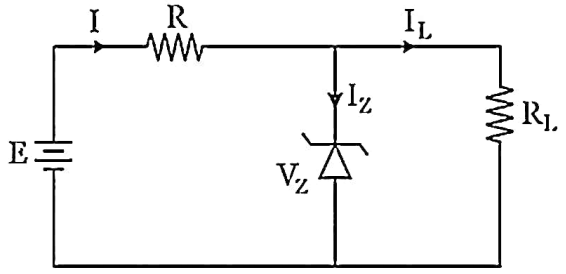
Question No. 9 **(Only One Option Correct Type)**

A hydrogen sample is prepared in a particular excited state A of quantum number, $n_A = 3$. The ground state energy of hydrogen atom is $-|E|$. The photons of energy $\frac{|E|}{12}$ are absorbed in the sample which results in the excitation of some electrons to excited state B of quantum number n_B, whose value is--

- 7
- 4
- 5
- 6

Question No. 10 **(Only One Option Correct Type)**

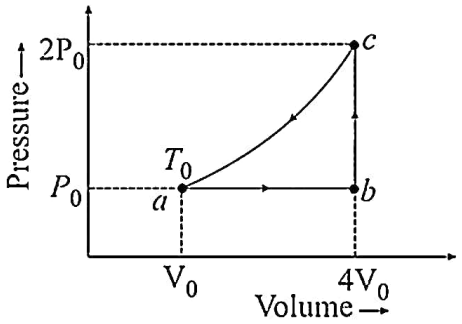
In the given circuit, if $R = 100\ \Omega$, $E = 5$ V and Zener diode is of 3 V, then the smallest value of load resistance for stabilization to be possible is



- 50 Ω
- 100 Ω
- 150 Ω
- 200 Ω

Question No. 11 **(Only One Option Correct Type)**

One mole of an ideal monoatomic gas is made to go through the cycle as shown in figure. Then the change in the internal energy in expanding the gas from a to c along the path abc is



- $3P_0V_0$
- $6RT_0$
- $4.5RT_0$
- $10.5RT_0$

Question No. 12 **(Only One Option Correct Type)**

A plane electromagnetic wave of frequency 50MHz travels in free space. If the average energy densities in the electric field and magnetic field are K_E and K_B respectively, then the correct option in the following is:

- $K_E = K_B$
- $K_E = K_B = 0$
- $K_E > K_B$
- $K_E < K_B$

Question No. 13 **(Only One Option Correct Type)**

Two fly wheels A and B are mounted side by side with frictionless bearings on a common shaft. Their moments of inertia about the shaft are 5.0 kg m^2 and 20.0 kg m^2 respectively. Wheel A is made to rotate at 10 revolution per second. Wheel B, initially stationary, is now coupled to A with the help of a clutch. The rotation speed of the wheels will become....

- $2\sqrt{5}$ rps
- 0.5 rps
- 2 rps
- 3 rps

Question No. 14 **(Only One Option Correct Type)**

Two resistances of 400 Ω and 800 Ω are connected in series with 6 volt battery of negligible internal resistance. A voltmeter of resistance 10,000 Ω is used to measure the potential difference across 400 Ω. The error in the measurement of potential difference in volt approximately is.

○ 0.01

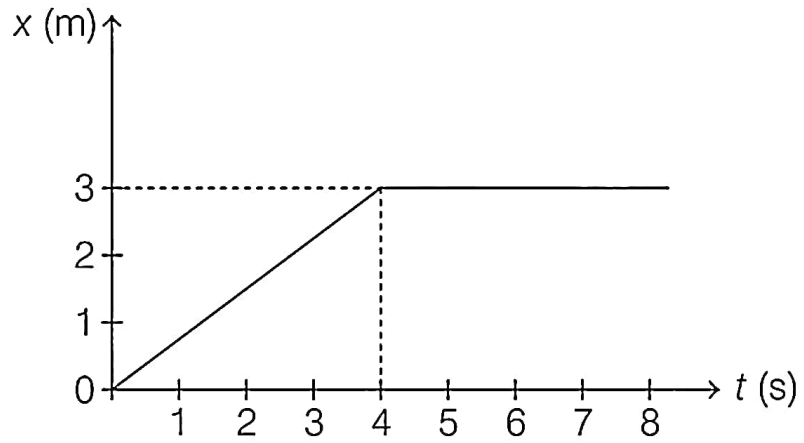
○ 0.02

○ 0.03

○ 0.05

Question No. 15 **(Only One Option Correct Type)**

The position-time $(x - t)$ graph of a moving body of mass 2 kg is shown in the figure. The impulse on the body at $t = 4$ s is



○ $1.5\ \text{kg} - \text{ms}^{-1}$

○ $-1.5\ \text{kg} - \text{ms}^{-1}$

○ $1\ \text{kg} - \text{ms}^{-1}$

D) -1 kg -ms^{-2}

Question No. 16 **(Only One Option Correct Type)**

Energy (E), angular momentum (L) and universal gravitational constant (G) are chosen as fundamental quantities. The dimensions of universal gravitational constant in the dimensional formula of Planck's constant (h) is-

○ zero

○ -1

○ $\frac{5}{3}$

○ 1

Question No. 17 **(Only One Option Correct Type)**

Two bars A and B of circular cross-section and of same volume and made of the same material are subjected to tension. If the diameter of A is half that of B and if the force applied to both the rods is the same and it is in the elastic limit, the ratio of extension of A to that of B will be-

○ 16 : 1

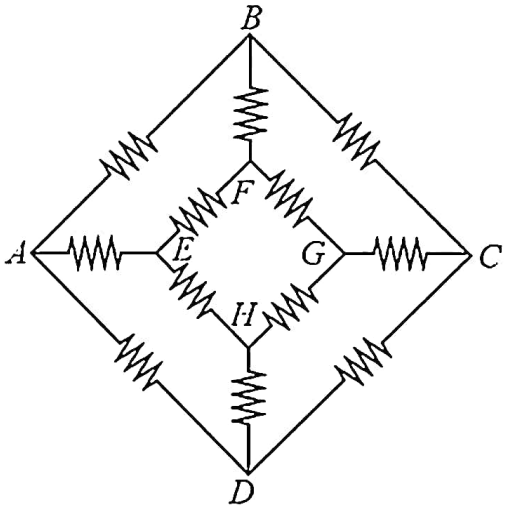
○ 8 : 1

○ 4 : 1

○ 2 : 1

Question No. 18 **(Only One Option Correct Type)**

In the network shown in figure, each resistance is R.



The equivalent resistance between A and C is-

○ $\frac{3R}{4}$

○ $\frac{5R}{6}$

○ $\frac{7R}{12}$

○ $\frac{4R}{3}$

Question No. 19 **(Only One Option Correct Type)**

A gas of mass ' m ' and molecular weight ' M ' is flowing in an insulated tube with a velocity ' 2 V '. If the flow of the gas is suddenly stopped and all the kinetic energy is utilized to compress the gas, the increases in the temperature of the gas is (γ is ratio of specific heats, R is universal gas constant)....

○ $\frac{2MV^2(\gamma-1)}{R}$

○ $\frac{mV^2(\gamma-1)}{2MR}$

○ $\frac{mV^2\gamma}{2R}$

○ $\frac{MV^2\gamma}{2R}$

Question No. 20 **(Only One Option Correct Type)**

Two wires A and B are of lengths 40 cm and 30 cm. A is bent into a circle of radius r and B into an arc of radius r. A current i_1 is passed through A and i_2 through B. To have the same magnetic inductions at the centre, the ratio of $i_1 : i_2$ is____

○ 3 : 4

○ 3 : 5

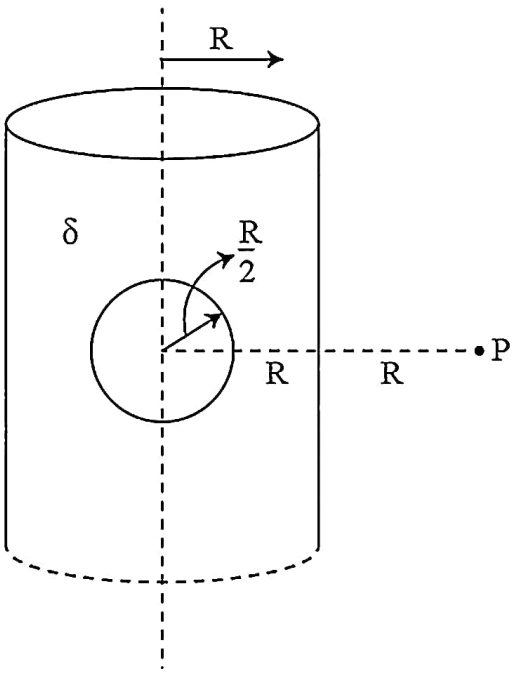
○ 2 : 3

○ 4 : 3

Physics Numerical (Maximum Marks: 20)

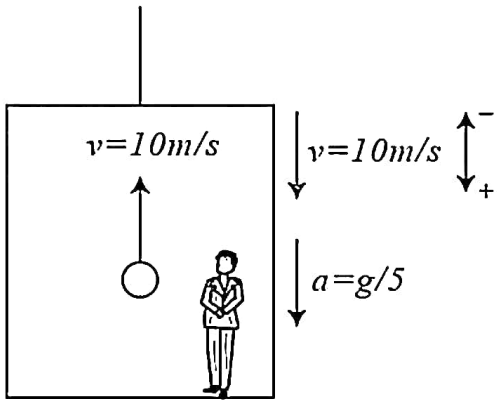
Question No. 1 **(Numerical Type)**

An infinitely long solid cylinder of radius R with uniform volume charge density ρ has a spherical cavity of radius $\frac{R}{2}$ with its centre on the axis of the cylinder as shown in the figure. The magnitude of the electric field at a point P which is at a distance $2R$ from the axis of the cylinder is given by $\frac{23\rho R}{6K\epsilon_0}$. What is the value of K?



Question No. 2 **(Numerical Type)**

A boy is in a lift which is moving downwards with an acceleration $g/5$. At an instant when the velocity of the lift is $10\ \mathrm{m/s}$, then boy throws a ball upwards with a velocity $10\ \mathrm{m/s}$. The time after which ball will return into the hand of boy is (take $g = 10\ m/s^2$)



Question No. 3 **(Numerical Type)**

YDSE is conducted using light of wavelength $6000\ \mathring{\mathrm{A}}$ to observe an interference pattern. When a film of some material 3.0×10^{-3} cm thick was placed over one of the slits, the fringe pattern shifted by a distance equal to 10 fringe widths. If the refractive index of matrial of the film is K, find 10K.

Question No. 4 **(Numerical Type)**

An open cylindrical vessel filled with water is rotated together with water at angular speed $1\mathrm{rad\ s^{-1}}$. At this angular speed, the centre of the bottom is just exposed. If the angular speed is increased to $\sqrt{\frac{3}{2}}\ \mathrm{rad\ s^{-1}}$, how much area (in $\mathrm{cm^2}$) of the bottom of the vessel is now exposed to surrounding? [Area of the cross-section of cylinder is $27\ \mathrm{cm^2}$]

Question No. 5 **(Numerical Type)**

A block of mass $2\ \mathrm{kg}$ is kept at the origin at $t = 0$ and is having velocity $4\sqrt{5}\ \mathrm{m\ s^{-1}}$ in the positive x direction. The only force on it is a conservative and its potential energy is defined as $U = -x^3 + 6x^2 + 15$.
Its velocity when the force acting on it is zero (after the time $t = 0$) is - (mark answer to nearest integer)

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1 **(Only One Option Correct Type)**

Deficiency of which vitamin causes Pernicious anaemia?

- Vitamin K
- Vitamin A
- Vitamin C
- Vitamin B_{12}

Question No. 2 **(Only One Option Correct Type)**

Find the final product ' E'

$$CH_2OH - CH = CH_2 \xrightarrow[\text{(ii) } NaSO_3H]{OsO_4} (A) \xrightarrow{KHSO_4} (B) \xrightarrow{Zn-Hg/\text{ conc. HCl}} (C) \xrightarrow{NBS/hv} (D) \xrightarrow{(CH_3)_2CuLi_4} (E)$$

- But-2-ene
- But-1-ene
- Isopentene
- Pen-2-ene

Question No. 3 **(Only One Option Correct Type)**

What will be the volume of O_2 Liberated at NTP by passing 5 A current For 193 Sec. through acidified water.

- 56 mL
- 112 mL
- 158 mL
- 965 mL

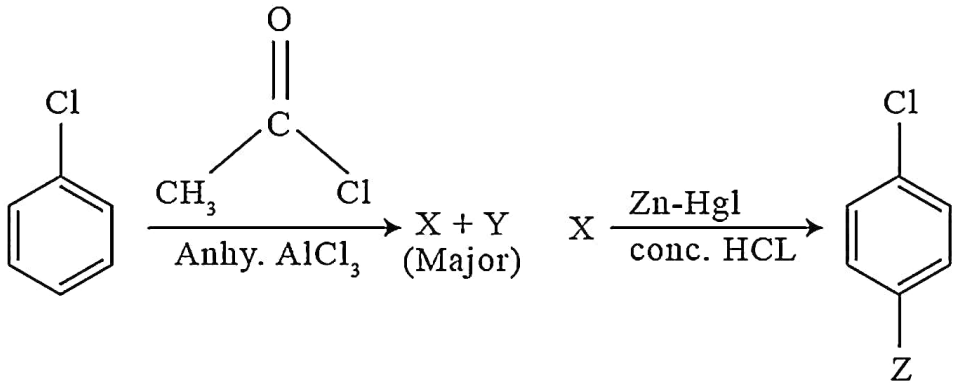
Question No. 4 **(Only One Option Correct Type)**

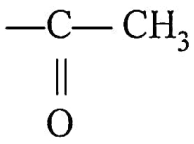
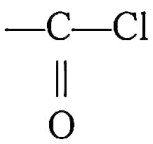
Which of the following statement is incorrect about the collision theory of chemical reaction?

- It considers reacting molecules or atoms to be hard spheres and ignores their structural features.
- Number of effective collisions determines the rate of reaction.
- Colliision of atoms or molecules possessing sufficient threshold energy results into the product formation.
- Molecules should collide with sufficient threshold energy and proper orientation for the collision to be effective.

Question No. 5 **(Only One Option Correct Type)**

What is Z in the following sequence of reactions?



- 
- 
- $-CH_3$
- $-CH_2 - CH_3$

Question No. 6 **(Only One Option Correct Type)**

Statement I : Glucose gives a reddish-brown precipitate with Fehling's solution.
Statement II : Reaction of glucose with Fehling's solution gives CuO and gluconic acid.

○ Statement I is true, Statement II is true, Statement II is a correct explanation for Statement I.

○ Statement I is true, Statement II is true, Statement II is not a correct explanation for Statement I.

○ Statement I is true, Statement II is false.

○ Statement I is false, Statement II is true.

Question No. 7 **(Only One Option Correct Type)**

Assertion : Given x mL of a sodium bicarbonate ($NaHCO_3$) solution requires y mL of hydrochloric acid (HCl), for complete neutralization. x mL of the same solution would require, for complete neutralization, y mL of sodium hydroxide (NaOH), whose molarity is same as that of hydrochloric acid (HCl).
Reason : sodium bicarbonate($NaHCO_3$) as an amphoteric salt that can be neutralized by either a strong acid or a strong base.

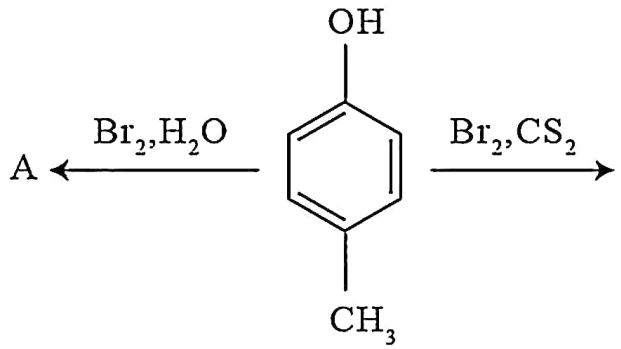
○ Both assertion and reason are correct, but reason is not the correct explanation of assertion.

○ Assertion is correct, but reason is wrong.

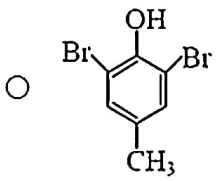
○ Assertion is wrong, but reason is correct.

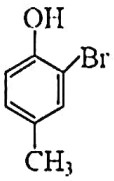
○ Both assertion and reason are correct and reason is the correct explanation of the assertion.

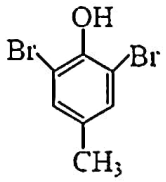
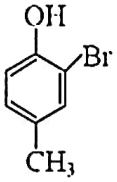
Question No. 8 **(Only One Option Correct Type)**

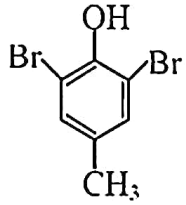
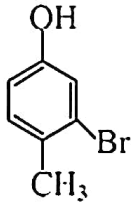


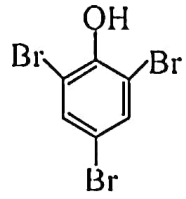
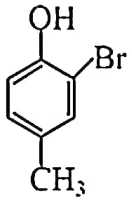
B Predominant 'A' and 'B' are respectively

○ 



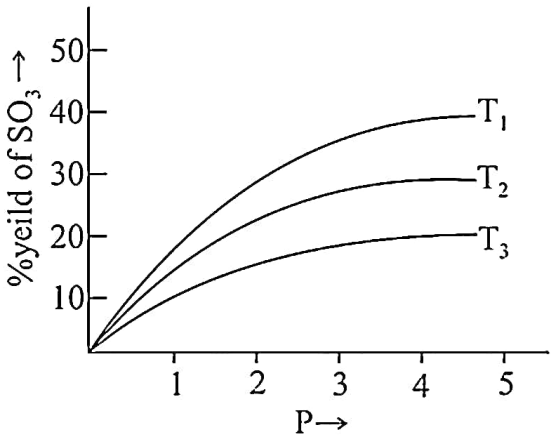
○  

○  

○  

Question No. 9 **(Only One Option Correct Type)**

The preparation of $SO_3(g)$ by the reaction, $SO_2\left(g\right) + \frac{1}{2}\,O_2\left(g\right) \rightleftharpoons SO_3\left(g\right)$,



shows the following figure in % yield against pressure at different temperature, then:

○ $T_3 > T_2 > T_1$

○ $T_1 > T_2 > T_3$

○ $T_1 = T_2 = T_3$

○ $T_2 > T_1 > T_3$

Question No. 10 **(Only One Option Correct Type)**

The uncertainties in the velocities of two particles A and B are 0.05 and 0.02 m/s^{-1} respectively. The mass of B is five times to that of mass A. What is the ratio of uncertainties $\left(\frac{\Delta x_A}{\Delta x_B}\right)$ in their positions?

○ 2

○ 0.25

○ 4

○ 1

Question No. 11 **(Only One Option Correct Type)**

Assertion (A) Covalent character follows $NaCl < MgCl_2 < BeCl_2 < LiCl$.

Reason (R) According to Fajan's rules, covalent character of ionic compounds increases with decrease in size of cation.

○ Both A and R are true and R is a correct explanation for A.

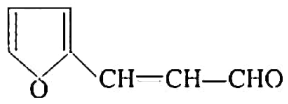
○ Both A and R are true but R is not a correct explanation for A.

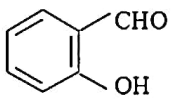
○ A is true but R is false.

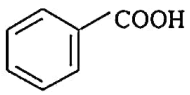
○ A is false but R is true.

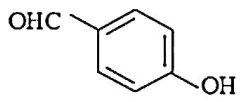
Question No. 12 **(Only One Option Correct Type)**

An aromatic compound 'P' having molecular formula $C_7H_6O_2$ on treating with aqueous ammonia and heating forms compound 'Q'. The compound 'Q' on reaction with molecular bromine and potassium hydroxide provides compound 'R' having molecular formula C_6H_7N . The structure of compound 'P' is

○ 

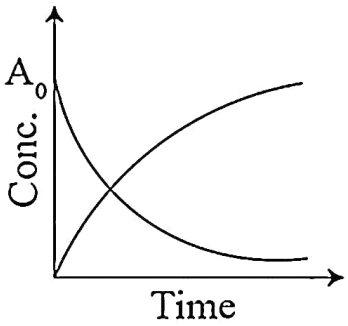
○ 

○ 

○ 

Question No. 13 **(Only One Option Correct Type)**

At the point of intersection of the two curves shown, the conc. of B is given by for, $A \rightarrow nB$



- $\frac{nA_0}{2}$
- $\frac{A_0}{n-1}$
- $\frac{nA_0}{n+1}$
- $\left(\frac{n-1}{n+1}\right)A_0$

Question No. 14 **(Only One Option Correct Type)**

Given below are two statement : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : Size of Bk^{3+} ion is less than Np^{3+} ion.

Reason R : The above is a consequence of the lanthanide contraction.

In the light of the above statements, choose the correct answer from the options given below :

- A is false but R is true
- Both A and R are true but R is not the correct explanation of A
- A is true but R is false
- Both A and R are true and R is the correct explanation of A

Question No. 15 **(Only One Option Correct Type)**

Consider the following complexes.

(i) $[FeIF(CN)(H_2O)(en)]$

(ii) $[MoCl_2F_2\ (gly)]^2$

Then, calculate value of $|x - y|$ (where x and y are total number of possible optically active isomers in (i) and (ii) complex respectively).

○ 15

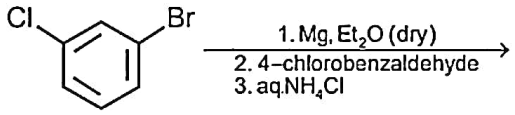
○ 12

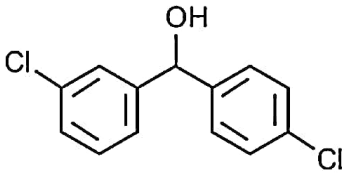
○ 8

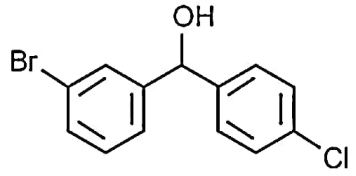
○ 6

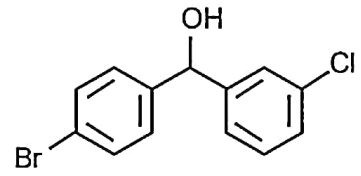
Question No. 16 **(Only One Option Correct Type)**

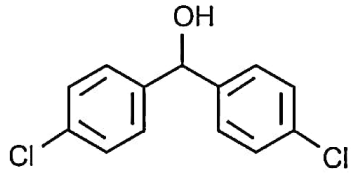
The product of the following reaction is :



○ 

○ 

○ 

○ 

Question No. 17 **(Only One Option Correct Type)**

A colourless water soluble solid 'X' on heating gives equimolar quantities of Y and Z. Y gives dense white fumes with HCl and Z does so with NH_3. Y gives brown precipitate with Nessler's reagent and Z gives white precipitate with nitrates of Ag^{+}, Pb^{2+} and Hg_2^{2+}. 'X' is :

○ NH_4Cl

○ NH_4NO_3

○ NH_4NO_2

○ $FeSO_4$

Question No. 18 **(Only One Option Correct Type)**

Which of the following isoelectronic ions has lowest ionisation energy?

○ Cl^-

○ Ca^{2+}

○ K^+

○ S^{2-}

Question No. 19 **(Only One Option Correct Type)**

Arrange the following bonds according to their average bond energies in descending order:
$C-Cl, C-Br, C-F, C-I$

○ $C-F > C-Cl > C-Br > C-I$

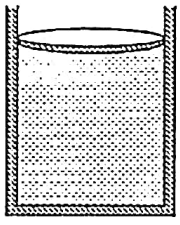
○ $C-Br > C-I > C-Cl > C-F$

○ $C-I > C-Br > C-Cl > C-F$

○ $C-Cl > C-Br > C-I > C-F$

Question No. 20 **(Only One Option Correct Type)**

An adiabatic cylinder has 8 gram of helium in it at 400 K and its piston of area 1 cm^2 (which has mass 1 kg) is movable. If a block of mass 62 kg is placed on the piston then the final temperature of the gas will be (Take atmospheric pressure to be $10^5 Nm^{-2}$ and $g = 10\ m/s^2$)



○ 500 K

○ 800 K

○ 1200 K

○ 1600 K

Chemistry Numerical (Maximum Marks: 20)

Question No. 1 **(Numerical Type)**

How many of the following compounds have $(p\pi - d\pi)$ multiple bonds?

i. SO_2
ii. SO_3
iii. $HSO_4^{\ominus}$
iv. SO_4^{2-}
v. SO_3^{2-}
vi. $HSO_3^{\ominus}$

Question No. 2 **(Numerical Type)**

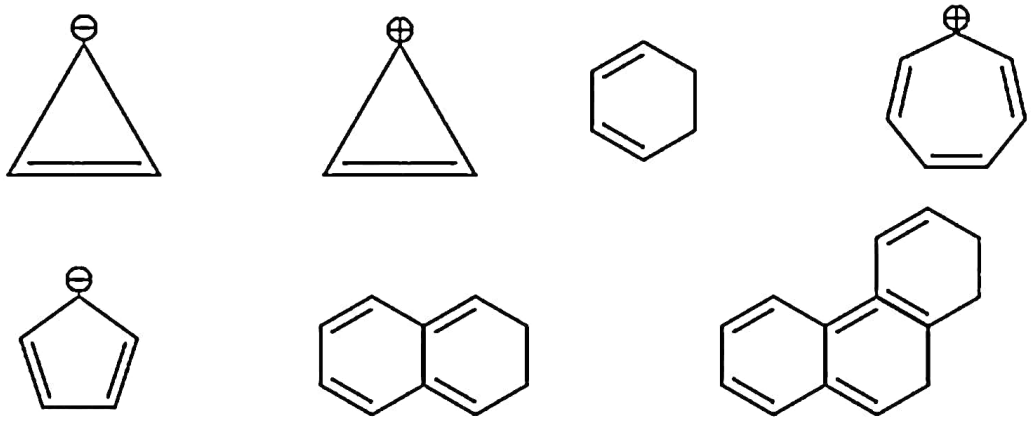
If the equilibrium constant of the reaction of weak acid HA with strong base is 10^9, then the pH of $0.1M$ NaA solution will be :

Question No. 3 **(Numerical Type)**

0.14 gram of $NaHCO_3$ tablet is added to neutralise 5 ml of 0.1 N HCl produced in stomach of a man. Then percentage purity of $NaHCO_3$ tablet is -

Question No. 4 **(Numerical Type)**

Among the following, the number of aromatic compounds are



Question No. 5 **(Numerical Type)**

For a dilute solution containing 2.5 g of a non-volatile non–electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is 2°C. Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (in mm of Hg) of the solution is (take $K_b = 0.76$ Kkg mol^{-1})

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1 **(Only One Option Correct Type)**

$ABCD$ is a quadrilateral. E is the point of intersection of the line joining the mid-points of the opposite sides. If O is any point and $\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}+\overrightarrow{OD}=x\overrightarrow{OE}$, then x is equal to

- 3
- 9
- 7
- 4

Question No. 2 **(Only One Option Correct Type)**

Let $f(x)$ be a function which is differentiable any number of times and $f\left(2x^2-1\right)=2x^3f\left(x\right),\ \forall\, x\in R$. Then $f^{(2010)}\left(0\right)=$

(Here, $f^{(n)}(x)=n^{\text{th}}$ order derivative of $f(x)$)

- -1
- 1
- 0
- data is insufficient

Question No. 3 **(Only One Option Correct Type)**

If $\mathrm{x}=\alpha$ satisfies the equation $\frac{\sin^{-1}x^2+\cos^{-1}x}{\cos^{-1}x^2+\sin^{-1}x}=-3$, then the value of $\frac{\sec^{-1}\alpha-\tan^{-1}\alpha}{\cot^{-1}\alpha-\operatorname{cosec}^{-1}\alpha}$ is

- 1
- 1.5
- 2
- 2.5

Question No. 4 **(Only One Option Correct Type)**

In a data with 15 number of observations $x_1, x_2, x_3, \ldots, x_{15}$, $\sum_{i=1}^{15} x_i^2 = 3600$ and $\sum_{i=1}^{15} x_i = 175$. If the value of one observation 20 was found wrong and was replaced by its correct value 40 , then the corrected variance of that data is

- 251
- 151
- 171
- 271

Question No. 5 **(Only One Option Correct Type)**

The particular solution of the different equation
$\frac{dx}{dy} = \frac{\sin y(1+y\cot y)}{x\log(xe^2)}, y(1) = 0$

- $y\sin y = x^2\log x$
- $y^2\sin y = \log x$
- $y = \left(\frac{e^2}{\sin e}\right)(x-1)$
- $y = e^2\sec x$

Question No. 6 **(Only One Option Correct Type)**

If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = -3\hat{i} + \hat{j} - \hat{k}$ and $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}, \vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ then a unit vector in the direction of $\overrightarrow{\mathrm{r}}$ is

- $\frac{1}{3}(-2\hat{i} + \hat{j} - \hat{k})$
- $\frac{1}{3}(-2\hat{\mathrm{i}} - \hat{\mathrm{j}} + 2\hat{\mathrm{k}})$
- $\frac{1}{3}(-2\hat{i} - \hat{j} - 2\hat{k})$
- None of these

Question No. 7 **(Only One Option Correct Type)**

If $f : [1, \infty) \to [\sqrt{2}, \infty)$ be a function defined by $f(x) = \sqrt{x^2 - 2x + 3}$ and g be inverse function of f then derivative of $g\left(f^2(x)\right)$ at $x = 3$ is:

- $\frac{6}{\sqrt{34}}$
- $\frac{\sqrt{34}}{6}$

- $\frac{12}{\sqrt{34}}$
- $\frac{24}{\sqrt{34}}$

Question No. 8 **(Only One Option Correct Type)**

Area bounded between the curves given by $|x| + y = 7$ and $4y = |4 - x^2|$, is (in sq. units)

- 24
- 36
- 32
- 28

Question No. 9 **(Only One Option Correct Type)**

The geometric mean of 6 observations was calculated as 13. It was later observed that one of the observations was recorded as 28 instead of 36. The correct geometric mean is

- $\left(\frac{9}{7}\right)^{\frac{1}{6}}$
- $3\left(\frac{9}{7}\right)^{\frac{1}{6}}$
- $13\left(\frac{9}{7}\right)^{\frac{1}{6}}$
- $13\left(\frac{7}{9}\right)^{\frac{1}{6}}$

Question No. 10 **(Only One Option Correct Type)**

I alternatively toss a fair coin and a fair die until I, either toss a head or throw a '2'. If I toss the coin first, the probability that I throw a '2' before I toss a head, is

- 1/7
- 7/12
- 5/12
- 5/7

Question No. 11 **(Only One Option Correct Type)**

Suppose a, b are integers and $a+b$ is a root of $x^2+ax+b=0$. What is the maximum possible values of b^2 ?

- 64
- 76
- 100
- None of these

Question No. 12 **(Only One Option Correct Type)**

Consider a polynomial $P(x)$ of the least degree that has a maximum equal to 6 at $x=1$ and a minimum equal to 2 at $x=3$. Then, the value of $P(2)+P'(0)-7$, is

- 4
- 3
- 6
- 8

Question No. 13 **(Only One Option Correct Type)**

Let R be a relation defined on the set Z of all integers and xRy, when $x+2y$ is divisible by 3 , then

- R is not transitive
- A is symmetric only
- R is an equivalence relation
- A is not an equivalence relation

Question No. 14 **(Only One Option Correct Type)**

Let $M\left(\frac{-7}{2}, \frac{-5}{2}\right)$ be the midpoint of the chord AB of the circle $x^2+y^2+10x+8y-23=0$. If $ax+$ by $1=0$ is the equation of AB then $3a+3b=$

- 1
- 1.5
- 3
- 2

Question No. 15 **(Only One Option Correct Type)**

If the line $\frac{x-2}{-1} = \frac{y+2}{1} = \frac{z+k}{4}$ is one of the angle bisectors of the lines $\frac{x}{1} = \frac{y}{-2} = \frac{z}{3}$ and $\frac{x}{-2} = \frac{y}{3} = \frac{z}{1}$, then the value of k is

- 1
- 2
- 4
- 8

Question No. 16 **(Only One Option Correct Type)**

$$A_1 = [a_1]$$

$$A_2 = \begin{bmatrix} a_2 & a_3 \\ a_4 & a_5 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} a_6 & a_7 & a_8 \\ a_9 & a_{10} & a_{11} \\ a_{12} & a_{13} & a_{14} \end{bmatrix} \ldots\ldots\ldots\ldots\ldots\ldots A_n = [\ldots\ldots\ldots]$$

Where $a_r = [\log_2 r]$ ([.] denotes greatest integer). Then trace of A_{10} is equal to

- 100
- 80
- 120
- 150

Question No. 17 **(Only One Option Correct Type)**

If the line joining the foci of the hyperbola $S_1 \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} + 1 = 0$ does not subtend a right angle at any point on the hyperbola $S_2 \equiv \frac{x^2}{4a^2} - \frac{y^2}{b^2} = 1$, then number of integral values of $4e^2$ is/are (e is eccentricity of $S_2 = 0$)

- 4
- 3
- 2
- 1

Question No. 18 **(Only One Option Correct Type)**

$A(z_1 = 2 + 2i), B(z_2), C(z_3)$ are three points on the Argand plane satisfying $|z_k - 2i| = 2, (k = 1, 2, 3)$. If $\triangle ABC$ encloses the maximum area, then the sum of the imaginary parts of z_2 and z_3 is

- 4
- 3
- 6
- 8

Question No. 19 **(Only One Option Correct Type)**

The value of the integral $\int_0^\infty \frac{\tan^{-1}x}{x^2+1+2x}\,dx$ is

- $\frac{\pi}{2}$
- $\frac{\pi}{8}$
- $\frac{\pi}{4}$
- $\frac{3\pi}{4}$

Question No. 20 **(Only One Option Correct Type)**

If $[x]$ is the greatest integer less than or equal to x and $|x|$ is the modulus of x, then the system of three equations

$2x + 3|y| + 5[z] = 0,$
$x + |y| - 2[z] = 4,$
$x + |y| + [z] = 1$ has

- a unique solution
- finitely many solutions
- infinitely many solutions
- no solution

Mathematics Numerical (Maximum Marks: 20)

Question No. 1 **(Numerical Type)**

Consider a function $f(x)$ which satisfies $f'(x) = \tan^2 x + \int_{\frac{-\pi}{4}}^{\frac{\pi}{4}} f(x)dx$. Also $f\left(\frac{\pi}{4}\right) = \frac{-\pi}{4}$.

If the value of $\frac{8+\pi^2}{\pi} \int_{\frac{-\pi}{4}}^{\frac{\pi}{4}} f(x)dx = m$, then find the value of m^2.

Question No. 2 **(Numerical Type)**

Coefficient of x^5 in $\frac{(1-x^2)^3(1-x^3)(1-x^4)}{(1-x)^5}$ is -

Question No. 3 **(Numerical Type)**

The first term of a sequence is 2014 . Each succeeding term is the sum of the cubes of the digits of the previous term. Then the 2014th term of the sequence is

Question No. 4 **(Numerical Type)**

If the number of ways of selecting two distinct numbers from the set {1, 2, ... , 12} whose sum is divisible by 3 is equal to k, then k - 16 is equal to .....

Question No. 5 **(Numerical Type)**

If the triangle ABC whose vertices are $A(-1, 1, 1), B(1, -1, 1)$ and $C(1, 1, -1)$ is projected on XY-plane, then the area of the projected triangle is

(ANSWER KEY)

Physics

Q1 [ ]	Q2 [ ]	Q3 [ ]	Q4 [ ]	Q5 [ ]
Q6 [ ]	Q7 [ ]	Q8 [ ]	Q9 [ ]	Q10 [ ]
Q11[ ]	Q12 [ ]	Q13 [ ]	Q14 [ ]	Q15 [ ]
Q16[ ]	Q17 [ ]	Q18 [ ]	Q19 [ ]	Q20 [ ]

Q21 [ ]	Q22 [ ]	Q23 [ ]	Q24 [ ]	Q25 [ ]

Chemistry

Q1 [ ]	Q2 [ ]	Q3 [ ]	Q4 [ ]	Q5 [ ]
Q6 [ ]	Q7 [ ]	Q8 [ ]	Q9 [ ]	Q10 [ ]
Q11[ ]	Q12 [ ]	Q13 [ ]	Q14 [ ]	Q15 [ ]
Q16[ ]	Q17 [ ]	Q18 [ ]	Q19 [ ]	Q20 [ ]

Q21 [ ]	Q22 [ ]	Q23 [ ]	Q24 [ ]	Q25 [ ]

Mathematics

Q1 [ ]	Q2 [ ]	Q3 [ ]	Q4 [ ]	Q5 [ ]
Q6 [ ]	Q7 [ ]	Q8 [ ]	Q9 [ ]	Q10 [ ]
Q11[ ]	Q12 [ ]	Q13 [ ]	Q14 [ ]	Q15 [ ]
Q16[ ]	Q17 [ ]	Q18 [ ]	Q19 [ ]	Q20 [ ]

Q21 [ ]	Q22 [ ]	Q23 [ ]	Q24 [ ]	Q25 [ ]